

# Which Microduck are we reviewing?

## 我们在比较哪一种Microduck?

**Our current version still uses fifteen XL330-M288-T servos.**

No cheaper motor has been substituted. The community replica and the commercial original are separate references; none of these records proves our version is a 1:1 copy.

当前版本仍采用**15个XL330-M288-T舵机**，没有换成廉价舵机。社区复刻和商业原版是两个不同参考；这些资料不能证明我们的版本是1:1复制。

| Topic / 项目       | Our ce-cad version / 我们的版本                                                                                                                   | Community replica / 社区复刻                                                                                                                      | Commercial original / 商业原版                                                                                                                                                            |
|------------------|----------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| What it is / 是什么 | Our engineering reconstruction in ce-cad, combining source meshes with rebuilt parts. Not certified 1:1. / 我们的ce-cad工程重建，混合参考网格与重建零件，未证明1:1。 | Independent community reconstruction by fanhao375. Public design/research files do not mean a completed robot. / fanhao375独立复刻研究，公开文件不代表整机完成。 | Pollen Robotics commercial robot. Software and some electronics are public; this is not a complete publicly released manufacturing definition. / Pollen商业机器人，软件及部分电子资料公开，但不是完整公开制造定义。 |

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|---------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Servos / 舵机               | 15 × ROBOTIS Dynamixel XL330-M288-T in our current model and parts list. / 当前模型和清单采用15个XL330-M288-T。                                                                                 | Baseline BOM specifies 15 × XL330-M288-T. HD-1910-C001 is a separate Feetech alternative under discussion. / 基线清单为15个XL330；飞特HD-1910-C001是另一路替代方案。                                         | Official runtime README says fifteen servos; public simulation supports the XL330-family inference. Exact fitted motor variant in the reviewed commercial unit remains unconfirmed. / 官方运行时说明15个舵机，公开模型支持XL330系列推断；目标实物具体型号未确认。 |
| Mechanics and fit / 机械与装配 | Reconstructed geometry and mesh repairs must retain evidence and new revisions. Assembly, interference and strength coverage are still incomplete. / 重建和网格修复需保留证据及版本；装配、干涉、强度尚未全部验证。 | Derived from public simulation meshes/MJCF. Printed shells, legs and feet and some leg screw fit are documented; that is not proof of every interface. / 来自公开仿真网格，已记录部分打印和腿部螺钉试装，并非所有接口验证。 | Manufacturer demonstrates working robots. Published simulation geometry does not certify original manufacturing tolerances, fasteners or the exact production revision. / 厂商展示了工作实机；公开仿真几何不代表生产公差、紧固件或具体版本。                     |

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|------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Electronics / 电子             | Public HAT C1 source is pinned, but our model differs from that board revision. Our historical HAT/IMU reconstructions have unresolved checks. / 公开HAT C1来源已固定，但当前网格板卡版本不同；旧HAT/IMU重建仍有未解决检查项。 | Public HAT reference plus a separate self-designed IMU. Its README explicitly says no prototype or physical verification yet. / 参考公开HAT，另自行设计IMU；作者明确说明IMU未打样、未实物验证。                               | Working electronics exist in the product; public HAT sources alone do not establish the installed HAT, IMU, camera or ToF revision. / 产品有工作电子系统，但公开HAT文件不能确定目标实机所有板卡版本。    |
| Software and control / 软件与控制 | ce-cad engineering review/simulation and archived upstream control sources. No completed walking validation on our physical robot. / ce-cad工程审查和仿真，加上上游控制资料；我们的实机尚未完成行走验证。                     | Studies/reuses upstream control architecture. Alternative servos require motor protocol, calibration and policy validation; source compatibility is not hardware proof. / 研究复用上游架构；换舵机需验证协议、标定和策略。 | Official Rust runtime drives fifteen servos at 50 Hz; official RL repository uses MuJoCo/PPO and exports ONNX policies. / 官方Rust运行时以50 Hz控制15个舵机，MuJoCo/PPO训练并导出ONNX策略。    |
| Physical evidence / 实物证据     | Review files and software checks are available; they do not establish a mechanically complete, powered and walking replica. / 有审查文件和软件检查，但不代表整机已完成、可通电行走。                                      | Printed-part and limited fit evidence; the pinned sources do not prove a complete walking replica. / 有打印及有限装配证据，所固定资料不能证明整机行走。                                                                     | Manufacturer videos demonstrate walking and other behaviors. We have not independently tested the exact production unit being compared. / 厂商视频展示行走等动作；我们没有独立测试此处对应的具体出厂实机。 |

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|------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Version control / 版本管理 | Project Git history plus immutable numbered reviews and file hashes. A working-tree render is not automatically a frozen release. / Git历史加不可变审查版本和文件哈希；工作区渲染并非自动冻结发布。 | Git checkout pinned to a1f4979c425b. Older mechanical archive 2549257 is separately retained. / 固定Git提交，旧机械归档2549257另行保留。 | Official runtime, RL and HAT repositories have Git histories. Those commits do not identify a physical product serial number or full production BOM. / 官方软件和HAT有Git历史，但提交不等于实物序列号或完整生产BOM。 |

## The cheaper-servo route / 更便宜舵机的路线

**Feetech HD-1910-C001 is a candidate, not a drop-in replacement or a purchase decision.** It needs its own reviewed revision. Validate mounting holes, horn/spline and clearances; supply voltage, connectors and pin order; bus protocol and driver; torque-speed behavior, backlash and calibration; and walking-policy compatibility. A fitting outline or similar three-wire cable does not settle these questions.

**飞特HD-1910-C001是候选方案，不是已确认可直接替换的器件，也不是采购决定。**需要单独审查版本，验证安装孔、舵盘与间隙、供电、接头针序、总线协议和驱动、力矩速度与回差标定，以及行走策略适配。

The XL330 supply mismatch already identified in our cable guide remains unresolved. Changing brands does not by itself close the electrical or mechanical checks. 当前XL330供电矛盾仍未解决；换品牌本身不等于通过电气和机械验证。

## Evidence and exact versions / 证据与版本

[Local source hashes and comparison record / 本地来源哈希与比较记录](#). The runtime and RL statements are supported by archived official READMEs, whose hashes are recorded; only commits actually recorded in the source manifests are claimed as pins.

- [Official runtime / 官方运行时](#) and [official training repository / 官方训练仓库](#).
- [Official HAT pin / 官方HAT固定版本](#).
- [Community replica at a1f4979c425b / 社区复刻固定版本](#).

Original fitted-servo identity: CANNOT DETERMINE beyond the stated source scope. Prices and component availability are not compared here.  
原版实物舵机的具体型号仍待确认；本页不比较价格与库存。